



Ratio

Teacher resource

Question 1

One thousand people attended a concert.
450 of the people were male and the rest were female.

What was the ratio of males to females at the concert?

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 9:11 | 9:10 | 9:20 | 9:13 |
| ☐ | ☐ | ☐ | ☐ |

Answer: A

Skill: Students apply knowledge of ratio.

Additional questions

1. What is the proportion of females at the concert?
2. What is the percentage of males at the concert?
3. How many extra females would have needed to attend the concert so that the ratio of males to females was 2:3?

Question 2

A recipe for Chocolate Crackles needs chocolate, golden syrup, butter and rice bubbles.

250 g chocolate
50 g golden syrup
50 g butter
50 g rice bubbles

The ratio by mass of **chocolate** to **golden syrup** to **butter** to **rice bubbles** is **5:1:1:1**.
What percentage by mass of the mixture is rice bubbles?

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 8% | 12.5% | 20% | 50% |
| ☐ | ☐ | ☐ | ☐ |

Answer: B

Skill: Students use knowledge of ratio and percentage in a familiar context.

Additional questions

1. To keep the same ratio as the above recipe, how much chocolate do I need if I use 150 grams of butter?
2. What percentage by mass of the mixture is chocolate?
3. What mass of golden syrup is in the mixture if the total mass of a mixture for chocolate crackles is 600 g?

Question 3

A garden centre makes its own potting mix made up of soil, compost and sand.

Soil makes up $\frac{2}{3}$ of the mix and compost makes up $\frac{1}{4}$ of the mix.

The remainder of the mix is sand.

What is the ratio of soil : compost : sand?

Answer: 8:3:1

Skill: Students use links between fractions and ratios.

Additional questions

1. What proportion of the potting mix is soil?
2. What percentage of the potting mix is compost?
3. The garden centre decides to change the mix so that the fraction of compost is reduced to $\frac{1}{6}$. Soil makes up $\frac{2}{3}$ of the mix and the remainder is sand.

What is the ratio of soil : compost : sand in this mix?

Question 4

Three friends shared the cost of a lottery ticket. They won \$720. They chose to share the money based on how much each put towards the cost of the ticket.

Name	Amount paid towards ticket (dollars)
Yu	2
Pete	3
Tai	4

Calculate the amount of prize money Tai received.

dollars

Answer: 320

Skill: Students use ratios to solve problems.

Additional questions

1. How does the amount that Yu won compare to the amount that Tai won?
2. How does the amount that Pete won compare to the amount that Tai won?
3. How does the amount that Pete won compare to the amount that Yu won?
4. How does the amount that Yu won compare to the amount that Pete won?

Student activity: Ratio

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